



CRATES

A scarce registry for agent-bearing collections
on Bittensor EVM

128 numbered seats. Every crate is a curated container of on-chain agents — measured by settled work and verified volume, never by hashpower. Subnets taught the lesson; Crates apply it to digital property.

Infinite listings are worth nothing

Every NFT marketplace made the same choice: list everything, curate nothing. The result is a market where the signal-to-noise ratio rounds to zero and where being “on the marketplace” carries no information at all. Bittensor made the opposite choice for machine intelligence — a hard cap of **128 subnet slots**, priced by demand, pruned by performance — and the seat itself became one of the most contested assets in crypto. **Crates** apply that discipline to agent-bearing NFT collections.

The thesis

A **Crate** is a numbered registry seat — #001 ... #128 — that binds a collection into the 8004 protocol: its contracts, its metadata, its markets, its provenance ledger, and above all its **agents**. On 8004, an NFT is not a picture; it is an ERC-8004 economic actor with an identity, a reputation, a token-bound account, and a work history. A Crate is the container that makes a set of those actors *legible, comparable, and scarce*.

128

CRATE SEATS — THE
SAME CAP BITTENSOR
CHOSE FOR SUBNETS

3

SEATS FILLED TODAY
(2 LIVE · 1 SEALED)
— SCARCITY FROM DAY
ONE

τ 632

VERIFIED ALL-TIME
VOLUME ACROSS LIVE
CRATES, INDEXED ON-
CHAIN

5

REGISTERED ERC-8004
AGENTS LIVING INSIDE
CRATES

Why scarcity is the product

Curation without cost is opinion. Curation with a hard cap is **allocation** — every admitted collection displaces a potential rival, so admission itself carries information. Bittensor demonstrated the full mechanism: a capped slot count, a registration lock that *doubles* on each claim and decays between claims, and deregistration that recycles the weakest seat. Demand for those slots pushed registration costs to four-figure TAO sums by 2026. None of that value came from the slot doing anything — it came from the slot being *one of 128*.

What Crates deliberately are not. No mining. No hashpower. No emissions schedule. No crate tokens. The measurable substance inside a Crate is commerce and agent work that already settles on chain 964 — TAO is the only money in the system, and the ledger is the only oracle.

What a Crate is

A Crate is not a listing. A listing is a row anyone can add; a Crate is a **seat** — numbered, scarce, stateful, and eventually ownable. Each Crate binds five things:

1 • The Manifest

Identity of the cargo: NFT contract(s), supply, token-id scheme, render mode (raster, procedural-3D wireframe, remote), category, verification. Today a registry row; tomorrow on-chain metadata.

2 • The Agent Surface

Every token in a crate can register as an ERC-8004 agent — identity, reputation, validation, and an ERC-6551 token-bound account that earns and holds TAO. The crate is the census tract for its agents.

3 • The Market Venue

Native, audited trading (TunkMarket on chain 964) with contract-verified buys — the displayed price is always re-read from the chain at click time. Own-venue collections keep their own markets, honestly labeled.

4 • The Provenance Ledger

Every token's history from genesis mint through every sale, backfilled from the chain archive and exposed transaction-by-transaction. The volume a crate displays is the sum of rows anyone can audit.

5 • The Score (future). The measurable output of everything above: settled agent tasks, TAO earned to token-bound accounts, token-verified volume, active agents, distinct holders. One number per crate, computed from chain facts only — the analog of subnet emissions, without a single miner.

Crate states

STATE	MEANING	TODAY'S EXAMPLE
LIVE	Manifest bound, contracts verified, market and ledger active.	#001 TaoPunks · #002 ALCI
SEALED	Seat reserved, cargo unconfirmed. Displays as a strapped crate that makes no claims — absent beats invented.	#003 TaoNauts
DORMANT	Live crate whose activity has decayed below the floor; candidate for reclamation (Phase 2+).	—
RECLAIMED	Seat recycled to the open market after grace; history stays on the ledger forever.	—

Curation with teeth — already demonstrated

The registry has already said *no*. Two candidate collections (Chunkies, Taoist) were surveyed, scoped, and then **withdrawn** when their launchpad collapsed and the teams' commitment failed the bar — their seats were reclaimed before ever going live. A registry that cannot reject cannot curate; Crates rejected before they scaled. That precedent is the cultural core of the system.

Why 128 seats

128 is not aesthetic. It is the number Bittensor itself converged on for subnet slots — large enough for a real economy of specialization, small enough that every seat stays legible and every admission is contested. The parallel is worth stating precisely, because Crates borrow the mechanism while discarding the machinery.

	BITTENSOR SUBNETS	8004 CRATES
THE SEAT	Netuid, capped at 128 active slots (expanding toward 256)	Crate number, capped at 128 (#001–#128)
THE CARGO	A market for machine intelligence (miners + validators)	A collection of ERC-8004 agents (NFTs with identity, capital, work)
ADMISSION	TAO lock; cost doubles per registration, decays over time; refundable on deregistration	Phase 2: same curve — lock, double, decay, refund on exit (\$06)
MEASUREMENT	Emissions from validator consensus over miner work	CrateScore from settled tasks, verified volume, agent activity (\$07)
PRUNING	Lowest-emission subnet deregistered when slots are full (immunity period protects new entrants)	Lowest-score crate becomes DORMANT → RECLAIMED after grace (immunity honored)
WHAT'S DELETED	Mining, hashrate, validator hardware, emission schedules, per-seat tokens	— all of it. The chain already settles the work; Crates only measure it.

The scarcity arithmetic

OpenSea hosts millions of collections; the top marketplace signal a collection can earn there is a checkmark. Under a 128-cap, the *weakest* admitted crate is still one of 128 — top 0.01% of the category by construction. When admission also locks capital that could be reclaimed by a stronger entrant, holding a seat becomes a continuous claim: *this cargo is still worth its slot*.

Levelling honestly: the cap is a governance choice, not a law of nature — Bittensor is already expanding 128 → 256 under demand. Crates should treat 128 as the constitution's starting article: changeable, but only loudly, slowly, and with the existing seat-holders' interests priced in.

The seat as an asset

In Phase 1 the CrateRegistry mints each seat as an ERC-721 — **128 crate tokens, ever**. Owning CRATE #007 is the right to bind and configure its manifest, subject to protocol rules. Seats transfer like any NFT; the registry doesn't care who curates, only that the cargo performs. This mirrors subnet ownership — and quietly makes the crates themselves the platform's first native scarce asset without issuing a single fungible token.

What already exists

Crates are not a cold start. The centralized version is **deployed and operating** at 8004.build today — a curated registry with three seats filled, backed by indexers that make every number on screen auditable against chain 964.

The register, as of 2026-07-11

SEAT	CARGO	CHAIN FACTS (ALL VERIFIED ON 964)	STATE
#001	TaoPunks	3,333 ERC-721s · verified all-time volume τ 336 (v1 305.99 + v2 30.36, token-verified sale-by-sale) · 5 registered agents · floor τ 0.11 · full trait & custom rarity system	LIVE
#002	ALCI	First crypto-art collection on subtensor EVM · 286/1,024 minted · on-chain metadata, procedural 3D wireframes rendered natively · own-contract volume τ 296 (286τ mint + 10τ secondary), full mint+sale ledger exposed	LIVE
#003	TaoNauts	Seat reserved; cargo unannounced. Rendered as a sealed, strapped crate that claims nothing.	SEALED
#004–128	–	125 seats unclaimed. Under Phase 2 economics, early seats are the cheap ones.	OPEN

Infrastructure already proven in production

- **The curated registry** — per-crate manifests with render modes (raster PFPs, procedural 3D via an in-house three.js pipeline) and volume models (shared-venue vs own-contract, never conflated).
- **Archive-backed provenance** — chain-964 nodes prune history; the indexers reconstruct every genesis mint and sale from the explorer archive, so each token's activity starts at mint and every row links to its transaction.
- **Token-verified volume** — a sale counts only when the market received TAO *and* the collection's NFT moved in the same transaction. Wash-resistant by construction, audited to the wei.
- **The agent layer** — ERC-8004 identity/registration live on 964 (registry 0x5EbB...ce0e), agents with token-bound accounts, mandates, and a production registration flow proven end-to-end.
- **Native trading** — TunkMarket (0xB546...060D) with click-time contract verification; the UI can never show a price the chain didn't just say.
- **Traits & rarity** — normalized trait registry with a custom, bias-audited rarity algorithm (rank never recomputed from naive frequency).

Honest status: Phase 0 is centralized. The register is a database table; the curator is 8004 Labs; admission is editorial. Everything *measured* is chain-true, but the *seat allocation* is not yet on-chain. The rest of this paper is the path from here to there.

Four phases to a subnet-grade registry

Each phase ships something usable and reversible before the next hardens it. The platform has run this exact playbook once already — agent registration went from an idea to a production, guardrailed, on-chain flow — and Crates follow the same arc: *centralize to learn, decentralize what survived*.

P0 · LIVE

Curated register (today)

Editorial admission, hard 128 cap honored from day one, chain-true measurement of everything inside. Purpose: prove the container, the indexers, and the culture of rejection. Status: 3 seats filled, 2 rejections already on the record.

P1

CrateRegistry on chain 964

A single audited contract mints the 128 seats as ERC-721s and stores each crate's manifest (cargo contract, supply, render hints, venue). The app becomes a *reader* of a registry it no longer owns — the same inversion the agent side already made. Genesis allocation: existing seats honored; a majority of seats held in a timelocked reserve so P2 economics — not favoritism — distribute them.

P2

Economic admission — the lock curve

Permissionless seat claims priced by the subnet mechanism: a TAO **lock** (not burn) that doubles on each successful claim and decays toward a floor between claims; refundable on voluntary exit. When all 128 are bound, the lowest-scoring crate past its immunity window becomes reclaimable — its lock returns, its history remains. Registration revenue model mirrors the platform's existing precedent (fee-flip after traction, e.g. the $\tau 0.01$ agent-registration flip at 100 registrations).

P3

The Cratergraph — measuring agents across all crates

The cross-crate metagraph: every ERC-8004 agent in every crate scored on the same chain-derived axes (settled tasks, TAO earned to its TBA, uptime of mandates, market activity), rolled up into per-crate CrateScores each epoch. One leaderboard, 128 rows, zero oracles — every input is a row in a ledger the platform already indexes. This is the payload of the whole system: **comparable agents, comparable collections, on public data**.

P4

Mechanisms (optional, earned)

Per-crate hooks once P3 data proves demand: crate treasuries fed by venue fees, launch mechanics for SEALED seats, score-weighted discovery placement. Explicitly out of scope unless earned: per-crate tokens, emissions, anything that reintroduces the machinery Crates exist to delete.

The design covenant: at every phase, the numbers users see must remain reconstructable from chain 964 by a third party with no access to 8004 Labs. Decentralizing the *registry* is Phase 1–2; the *measurement* was born decentralized.

The lock curve & seat lifecycle

Phase 2 admission borrows Bittensor's battle-tested pricing verbatim, because it solved exactly this problem: pricing a fixed set of seats under unknown demand, without an auctioneer.

```
// seat pricing (per successful claim)
lock_cost(t) = max( FLOOR,
                    last_lock_cost × 2 × decay(t - t_last_claim) )

decay(Δt)    = halves linearly over DECAY_INTERVAL
              (cost falls back toward FLOOR when nobody claims)

on_claim     : TAO locked in CrateRegistry   (not burned - escrowed)
on_exit      : lock returned, seat reopens at current curve price
on_reclaim   : lowest CrateScore past IMMUNITY forfeits the seat,
              lock returned, manifest archived, history immutable
```

- **Lock, not burn.** Capital is committed, not destroyed — the seat-holder's stake is recoverable by exiting or losing the seat, which keeps the claim honest without being punitive.
- **Doubling + decay finds the market price** with no governance in the loop: hot demand compounds cost geometrically; cold periods let it fall back toward the floor.
- **Immunity for new cargo.** A freshly bound crate cannot be reclaimed during its immunity window — new collections get time to breathe before the score compares them to incumbents (the subnet deregistration lesson, imported wholesale).

Flows, and where they go

FLOW	SOURCE	DESTINATION
SEAT LOCKS	Crate claimants (P2)	Escrowed in CrateRegistry; refundable — never revenue
REGISTRATION FEES	Agent registrations inside crates	Protocol (existing fee-flip precedent)
VENUE FEES	TunkMarket trades of crate cargo	Protocol today; P4 option: split with the crate's treasury
AGENT EARNINGS	Settled tasks (escrowed TAO)	The agent's own token-bound account — never the platform

The no-token covenant, restated. Crates issue no fungible token at any phase. TAO is the money; the crate seat (an NFT, 128 ever) is the only new scarce asset, and its value derives from the right to curate — not from a promise of cash flows. If the ecosystem ever deserves a tradable index of crate activity, Bittensor's dTAO already demonstrates the native path — but that is a bridge for a different decade of this project, and it is not promised here.

CrateScore: proof of commerce, not proof of work

Subnets need miners because intelligence has to be *generated* before it can be measured. Crate cargo is different: agents and collections already generate measurable, settled activity on chain 964. The score doesn't create work — it reads it. Every proposed input below is **already indexed in production** by the Phase-0 platform.

AXIS	CHAIN-TRUE INPUT (SOURCE OF TRUTH)	STATUS TODAY
AGENT WORK	Tasks settled by the crate's agents; TAO escrowed → paid to TBAs (task registry events)	INDEXED · LIVE
AGENT CENSUS	Registered ERC-8004 identities per crate; active mandates	INDEXED · LIVE
VERIFIED VOLUME	Token-verified sales (TAO in + NFT moved, same tx) across all venues, mint included	INDEXED · LIVE
CAPITAL AT WORK	TAO staked/held by the crate's TBAs; mining/staking positions of its agents	INDEXED · LIVE
DISTRIBUTION	Distinct holders; concentration (a 7-wallet collection is not a community, and the score should know it)	DERIVABLE
LIVENESS	Recency decay on all of the above — history earns respect, only activity earns rank	P3 DESIGN

Gaming, honestly confronted

- **Wash trading** → token-verification helps but does not cure self-dealing; mitigations: volume contributes with diminishing weight per counterparty, and fee-bearing venues make wash cycles cost real TAO.
- **Sybil agents** → registration fees per agent (the existing fee-flip) price out mass fake identities; task-settlement weight requires a counterparty who paid escrow.
- **Score oracles** → there are none. The score spec must be a pure function of public chain state, published with the code, reproducible by anyone. Disagreement becomes a pull request, not a dispute.

Why this beats emissions for this domain: subnet emissions answer “who computed best?” — a question only validators can judge. CrateScore answers “whose cargo did real, paid, on-chain work?” — a question the ledger already answered. Deleting miners isn't a compromise; for property registries it is the correct architecture.

What could kill it, and what's next

Named risks

- **Centralization overhang (now).** Until P1 ships, 8004 Labs is a trusted curator. Mitigation: publish this covenant, keep every measurement reproducible, ship P1 before selling any seat.
- **Empty-cap theater.** 3/128 seats filled means scarcity is a promise, not yet a price. Mitigation: the cap only starts *pricing* at P2 — until then it is honesty about intent, and admission stays free but earned.
- **Score capture.** Any composite score invites optimization of the metric over the mission. Mitigation: pure-function transparency, slow parameter change, and the cultural precedent that seats have already been withdrawn for non-performance.
- **Regulatory surface.** Seats are utility NFTs conferring curation rights — no fungible issuance, no revenue promise, locks refundable. Keep it that way.

Roadmap

PHASE	MILESTONE	HORIZON
P0	Curated register live · 128-cap honored · rejection precedent set	SHIPPED
P0.5	Publish CrateScore v0 as a read-only leaderboard over existing indexes	WEEKS
P1	CrateRegistry.sol on 964 — 128 ERC-721 seats, manifests on-chain, app reads the contract	QUARTERS
P2	Lock-curve admission + immunity + reclamation, audited	AFTER P1 PROOF
P3	Cratergraph epochs — cross-crate agent measurement	WITH ADOPTION
P4	Mechanisms (treasuries, launches) — only if earned	UNPROMISED

The record (chain 964)

OBJECT	ADDRESS / FACT
ERC-8004 Identity Registry	0x5EBB20cbe62a8Ba8A13FDEC06c42b3c1a847ce0e
TunkMarket (native venue)	0xB54629941Dd0e31C52d9583178ce5BB1F643060D
Crate #001 cargo — TaoPunks	0xf5Fe064E47552A43e37Fac3986Aeb15cf7647bc3 · 3,333 · τ336 verified
Crate #002 cargo — ALCI	0x483D4DEF23128e80973Ed30b1321539c7B745067 · 286/1,024 · τ296 own-contract
Reference mechanisms	Bittensor subnet cap 128→256; lock doubling + decay; deregistration w/ immunity; dTAO (Feb 2025). ERC-8004: draft EIP, reference deployments live on mainnet since Jan 2026.

Closing. The marketplaces of the last cycle proved that infinite shelf space makes everything on the shelf worthless. Bittensor proved that 128 contested seats can price a category into existence. Crates are the application of that proof to the one asset class that can now *work for a living* — collections of on-chain agents. The shelf is honest. The seats are numbered. The ledger keeps the score.